

Module 1: Software Processes & Agile

Waterfall, V-Model, Prototyping, Boehm's Spiral 4 Quadrants & Scrum Ceremonies

☰ **Module I: Introduction & Process Models — Complete 14-Topic Syllabus Tracker**

- Topic 1:** What is Software Engineering? (Systematic Discipline)
- Topic 2:** Why Software Engineering? (Software Crisis & Complexity)
- Topic 3:** The Evolving Role of Software (Product, Service & System)
- Topic 4:** Software Process Framework (Core Generic Activities)
- Topic 5:** Classical Waterfall Model (Linear Sequential Lifecycle)
- Topic 6:** Prototyping Model (Requirement Discovery & User Feedback)
- Topic 7:** Boehm's Spiral Model (Risk-Driven Iterative Cycles)
- Topic 8:** Rapid Application Development (RAD Component Reuse)
- Topic 9:** Incremental Process Model (Progressive Feature Delivery)
- Topic 10:** Agile Models (Agile Manifesto & Sprint Feedback Loops)
- Topic 11:** Software Management Activities (The 4 P's Framework)
- Topic 12:** Project Planning (Scope, Cost & Resource Allocation)
- Topic 13:** Project Scheduling (WBS, Gantt Charts & PERT/CPM)
- Topic 14:** Risk Management (I-A-P-M-M Risk Assessment Cycle)

Topic 1 – 3: What is Software Engineering & The Software Crisis?

Software Engineering (IEEE 610.12) is the systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering principles to software.

Ad-Hoc Programming	Software Engineering Discipline
Individual developer writing code without formal specifications.	Multi-person, multi-team systematic engineering development lifecycle.
No formal documentation; maintainability ignored.	Comprehensive SRS, architectural designs, test plans, and version control.
Massive budget overruns, missed deadlines, high defect rates.	Predictable cost estimation (COCOMO), scheduling (PERT/Gantt), and QA/QC.

Topic 4 – 10: Software Development Process Models

Process Model	Core Lifecycle Mechanism & Characteristics	Ideal Project Suitability
1. Waterfall Model	Strict linear sequential phases (Requirements → Design → Code → Test → Deploy). Each phase produces sign-off deliverables before the next begins.	Well-understood, stable requirements with mature technology stacks (e.g., Space flight software, banking backends).
2. Prototyping	Builds an early disposable mock-up interface to elicit feedback, refine ambiguous user requirements, and discard/re-engineer.	High user interaction systems where clients have unclear or fuzzy initial requirements.
3. Spiral Model (Boehm)	Risk-driven iterative model structured in 4 quadrants: (1) Objective Determination, (2) Risk Analysis , (3) Engineering, (4) Customer Evaluation.	Large-scale, expensive, mission-critical, high-risk enterprise projects.
4. RAD Model	Rapid Application Development (60–90 days) utilizing component-based construction, GUI builders, and automated code generators.	Well-modularized information systems with short delivery deadlines and high component reuse.
5. Incremental Model	System is decomposed into modular release increments. Increment 1 delivers core functionality; subsequent releases add peripheral features.	Projects requiring early partial operational software release to capture market share.
6. Agile (Scrum / XP)	Iterative 2–4 week Sprints emphasizing working software over documentation, customer collaboration, and adaptive response to changing requirements.	Dynamic web/mobile products with volatile, continuously evolving market requirements.

Memory Hook: Process Model Selection Strategy

- Stable requirements → **Waterfall**
- Unclear user requirements → **Prototyping**
- High financial/technical risk → **Spiral**
- Rapid component delivery → **RAD / Incremental**
- Changing, dynamic requirements → **Agile**

Topic 11 – 14: Project Planning, Scheduling & Risk Management

The 5–Stage Risk Management Cycle (I–A–P–M–M)

1. **Risk Identification:** Identify project, product, and business risks (e.g., staff turnover, scope creep).
2. **Risk Analysis (Estimation):** Quantify Risk Probability (P) and Risk Impact / Severity (I). **Risk Exposure** (RE) = $P \times I$.
3. **Risk Prioritization:** Rank risks in a Risk Matrix to focus engineering resources on top hazards.
4. **Risk Mitigation (RMMM Plan):** Develop Risk Mitigation, Monitoring, and Management strategies to minimize impact.
5. **Risk Monitoring:** Continuously track risk indicators throughout the project lifecycle.

M1 Active Recall & Exam Questions

Q1. Compare Waterfall, Prototyping, Spiral, and Agile models across 5 engineering parameters. (10 Marks)

1. **Requirement Flexibility:** Waterfall is rigid; Prototyping clarifies fuzzy needs; Spiral adapts per cycle; Agile welcomes changes at any stage.
2. **Risk Management:** Spiral explicitly incorporates formal Risk Analysis in every cycle; Waterfall and Prototyping have zero built-in risk analysis; Agile manages risk via short sprint iterations.
3. **Customer Involvement:** Waterfall involves customer only at start and end; Agile and Prototyping require continuous customer collaboration.
4. **Delivery Schedule:** Waterfall delivers product only at the very end; Incremental/Agile delivers functional software increments every few weeks.
5. **Cost & Documentation:** Waterfall is documentation-heavy; Spiral is expensive due to risk experts; Agile prioritizes working software over extensive documentation.